Homework #4

Dungeon Master Part 1

Setup Tasks

- A Text Editor
- A Ubuntu Server
- Github Account
- The ability to push/pull/commit

Create a new public repository for this assignment, do this as the first step

Homework Description

For this assignment you're going to create a file called **dungeonmaster.c** which will read from a room description file (that will be provided) and store the information of each room into a structure. Once you're done reading through the file, output a brief description of the file contents like so:

Room <room code>: <room name>

See example output:

```
terratoast@terratoast-VirtualBox:~/Programs$ ./a.out dungeonRoomList.txt
Room FS: Forgotten Shrine
Room SH: The Serpent's Hall
Room AS: Arcane Study
Room IV: The Iron Vault
Room HM: Hall of Mirrors
Room SD: Spider's Den
Room BP: The Blood Pool
Room HA: The Hidden Armory
Room CT: Cursed Tomb
Room TC: The Trial Chamber
```

Example output for larger file:

```
terratoast@terratoast-VirtualBox:~/Programs$ ./a.out dungeonRoomList_large.txt
Room FS: Forgotten Shrine
Room SH: The Serpent's Hall
Room AS: Arcane Study
Room IV: The Iron Vault
Room HM: Hall of Mirrors
Room SD: Spider's Den
Room BP: The Blood Pool
Room HA: The Hidden Armory
Room CT: Cursed Tomb
Room TC: The Trial Chamber
Room AP: The Abyssal Pit
Room CC: The Crystal Cavern
Room SC: The Silent Crypt
Room IC: The Inferno Chamber
Room EC: The Echoing Cavern
Room PC: The Prison Cells
Room GW: The Golem Workshop
Room HC: The Hall of Chains
Room OC: The Obsidian Chamber
Room TD: The Torturer's Den
```

Some special notes about how you process the input:

- The program should not have prior knowledge of the number of rooms in the file. It will need to dynamically allocate space and expand the array.
- The strings that it reads into and saves in the structure should be trimmed of whitespace.
- The filename should be passed to the program via command line argument.
- While the description of the room is not required to be printed out for this assignment, double check to make sure that the description is being saved properly into the structure. This will be more important for a future assignment.

Rubric

Program Crashes	-4pt
Code not commented	-1pt
Done in one large commit	-1pt
Program reads from the file successfully (line by line)	2pts
Program creates array of room structures	2pts
Whitespace is trimmed from strings	2pts
Program takes the information from the file, and fills the rooms with them.	4pts
Program outputs the proper information (in the right format)	3pts
Filename retrieved from command line argument	2pt